
MATH 241 Engineering Mathematics I

Problem Set - III Solutions

Problem 1

Let w_1, w_2 be any matrices in W given as

$$w_1 = \begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} \in W, \quad w_2 = \begin{bmatrix} a_2 & b_2 \\ b_2 & c_2 \end{bmatrix} \in W$$

We need to satisfy the following 3 conditions to prove that W is a subspace.

1. $\underline{0} \in W$

$$\underline{0} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in W$$

2. $w_1 + w_2 \in W$

$$w_1 + w_2 = \begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} + \begin{bmatrix} a_2 & b_2 \\ b_2 & c_2 \end{bmatrix} = \begin{bmatrix} a_1 + a_2 & b_1 + b_2 \\ b_1 + b_2 & c_1 + c_2 \end{bmatrix} \in W$$

3. $\alpha \cdot w_1 \in W$

$$\alpha \cdot w_1 = \alpha \begin{bmatrix} a_1 & b_1 \\ b_1 & c_1 \end{bmatrix} = \begin{bmatrix} \alpha \cdot a_1 & \alpha \cdot b_1 \\ \alpha \cdot b_1 & \alpha \cdot c_1 \end{bmatrix} \in W$$

To verify that given matrices B_1, B_2, B_3 form a basis of W , we need to satisfy the following 2 conditions.

1. $\text{span}\{B_1, B_2, B_3\} = W$ Let a matrix $w \in W$ is written as,

$$w = \begin{bmatrix} a & b \\ b & c \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = a \cdot B_1 + b \cdot B_2 + c \cdot B_3$$

Hence, $w \in \text{span}\{B_1, B_2, B_3\} \Rightarrow W = \text{span}\{B_1, B_2, B_3\}$

2. Linear Independence: $a_1 \cdot B_1 + a_2 \cdot B_2 + a_3 \cdot B_3 = 0$ only if $a_1 = a_2 = a_3 = 0$

$$a_1 \cdot B_1 + a_2 \cdot B_2 + a_3 \cdot B_3 = a_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + a_3 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} a_1 & a_2 \\ a_2 & a_3 \end{bmatrix}$$

is 0 if all its entries are 0, hence, $a_1 = a_2 = a_3 = 0$. This implies linear independence, which verify these matrices to be basis of W .

Observe

$$\begin{bmatrix} 1 & -2 \\ -2 & 3 \end{bmatrix} = (1) \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + (-2) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + (3) \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = 1 \cdot B_1 - 2 \cdot B_2 + 3 \cdot B_3$$

Hence, the coordinates are (1,-2,3)

Problem 2

- (a) Let $B = \{b_1, b_2, \dots, b_m\}$ be a basis for S i.e., $\dim(S) = m$. since $B \subset S \subset V$ and B is linearly independent, V catches at least m linear independent vectors, which implies $\dim(V) \geq \dim(S) = m$.
- (b) Let $n = \dim(S) = \dim(V)$. Let a basis of S is $B = \{b_1, b_2, \dots, b_n\}$. since $B \subset S \subset V$ and B is a set of n linear independent vectors that are in V as well. since $\dim(V) = n$, these vectors must span V as well. That is, B is a basis of V as well, which implies $\dim(V) = \dim(S) = n$.
- (c) A basis for S is a set of independent elements in S . The basis for V is also set of independent elements in V . Then, if S spans V , it is a basis of V . If not, then there exist some vector y , that can be inserted to enlarge S and making $\{S, y\}$ a basis. This process continues until the enlarged S spans V . Hence, every basis of S is contained in some basis of V .
- (d) Let V be the vector space of polynomials of degree ≤ 2 and consider the basis $\{1, 1+x, x^2\}$ for V . Let S be the subspace consisting of polynomials of degree ≤ 1 , consider the basis $\{1, x\}$ for S . As $\{1, x\} \not\subset \{1, 1+x, x^2\}$, therefore, a basis for V need not contain a basis for S .

Problem 3

(a) We need to satisfy the following conditions to prove that W is a subspace.

i) $W = U + V \Rightarrow \underline{0} + \underline{0} = \underline{0} \in W$

ii) Let $\underline{u}_1 + \underline{v}_1, \underline{u}_2 + \underline{v}_2 \in W$ where $\underline{u}_1, \underline{u}_2 \in U$ and $\underline{v}_1, \underline{v}_2 \in V$ since U and V are subspaces, $\alpha\underline{u}_1 + \beta\underline{u}_2 \in U$ and $\alpha\underline{v}_1 + \beta\underline{v}_2 \in V$. Hence,

$$\alpha(\underline{u}_1 + \underline{v}_1) + \beta(\underline{u}_2 + \underline{v}_2) = (\alpha\underline{u}_1 + \beta\underline{u}_2) + (\alpha\underline{v}_1 + \beta\underline{v}_2) \in W.$$

Therefore, W is a subspace.

(b) No, it is not always unique. The condition $U \cap V = \{0\}$ must be satisfied. Let us first consider the case when $U \cap V \neq \{0\}$. Consider two subspaces

$$U = \left\{ \begin{bmatrix} a & b \\ b & c \end{bmatrix} \mid a, b, c \in \mathbb{R} \right\} \text{ and } V = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} \mid a, b \in \mathbb{R} \right\}$$

The $U \cap V$ are the matrices for which $c = a$.

Example:

$$\begin{aligned} W &= \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix} = U_1 + V_1 = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \\ W &= \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix} = U_2 + V_2 = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 4 & 1 \end{bmatrix} \end{aligned}$$

Let us now consider the case when $U \cap V = \{0\}$, then every $w \in W$ can be expressed in a unique way. Suppose, we can express $\underline{w}$ in two ways, given as

$$\underline{w} = \underline{u}_1 + \underline{v}_1 = \underline{u}_2 + \underline{v}_2,$$

hence

$$\underline{u}_1 - \underline{u}_2 = \underline{v}_2 - \underline{v}_1$$

As, U and V are subspaces, we have, $\underline{u}_1 - \underline{u}_2 \in U$ and $\underline{v}_2 - \underline{v}_1 \in V$. So, $\underline{u}_1 - \underline{u}_2 = \underline{v}_2 - \underline{v}_1 \in U \cap V$. As $U \cap V = \{0\}$, this means

$$\underline{u}_1 - \underline{u}_2 = \underline{v}_2 - \underline{v}_1 = 0 \Rightarrow \underline{u}_1 = \underline{u}_2, \underline{v}_1 = \underline{v}_2.$$

(c) Since U and V are subspaces of S , this means $p, q \leq n$. As W is also a subspace of S , hence, $\dim(W) \leq n$. W is formed by the addition of U and V , which implies the dimension of S to result in $p + q$. We have two conditions for this:

- i) $p + q \leq n$: As basis for U is p -dimensional $(u_1, \dots, u_p)$ and basis for V is q dimensional $(v_1, \dots, v_q)$, taking all the combination to generate basis of W results in dimension of W to be $p + q$ i.e., $(u_1 + v_1, u_1 + v_2, \dots, u_1 + v_p, u_2 + v_1, \dots, u_p + v_q)$. If $p + q \leq n$, the dimension of W will be $p + q$.
- ii) $p + q > n$: As by (i) dimension of W is $p + q$. If $p + q > n$, the dimension of W will be greater than n which contradicts the condition for subspace. Hence, dimension of W must be less than or equal to dimension of S . In this case the dimension of W is n .

Combining, (i) and (ii) we have, $\dim(W) = \min\{n, p + q\}$

Problem 4

This is false. Let $V = \mathbb{R}^2$ and let $U_1 = \text{span}(0, 1), U_2 = \text{span}(1, 1), W = \text{span}(1, 0)$. Geometrically, U_1, U_2, W are three mutually distinct lines. Then

$$U_1 + W = U_2 + W = \mathbb{R}^2 \text{ but } U_1 \neq U_2.$$

Problem 5

Since $W_1 \neq V$, $\exists \alpha_2 \in V$ such that $\alpha_2 \notin W_1$. If $\alpha_2 \notin W_2$ there is nothing to prove, so let $\alpha_2 \in W_2$. Since $W_2 \neq V$, $\exists \alpha_1 \in V$ such that $\alpha_1 \notin W_2$. If $\alpha_1 \notin W_1$, there is nothing to prove, so let $\alpha_1 \in W_1$. Then consider $\alpha = \alpha_1 + \alpha_2 \in V$

- $\alpha \notin W_1$ because if $\alpha \in W_1$, then $\alpha - \alpha_1 = \alpha_2 \in W_1$, contradiction.
- $\alpha \notin W_2$ because if $\alpha \in W_2$ then $\alpha - \alpha_2 = \alpha_1 \in W_2$, where $\alpha \in W_2$ and $\alpha_2 \in W_2$, contradiction.

Therefore, $\alpha \in V$ such that $\alpha \notin W_1$ and $\alpha \notin W_2$. To show that V has a basis that contains no vectors in W_1 and W_2 , let $W_3 = \text{span}(\alpha)$. We claim that there exist a vector β that is not contained in any of W_1, W_2, W_3 . To see this, pick $w_3 \notin W_3$ and consider $\beta_1 = \alpha + w_3, \beta_2 = \alpha + 2w_3$, and $\beta_3 = \alpha + 3w_3$. If they all fell in $W_1 \cup W_2$, then at least two would be in W_1 or W_2 , say, β_1 and β_3 in W_2 . It is immediate that $3\beta_1 - \beta_3 = 2\alpha \in W_2$ and $\alpha \in W_2$, a *contradiction*. Let $\beta \notin W_1 \cup W_2 \cup W_3$. Then α and β are linearly independent. Now, put $W_4 = \text{span}(\alpha, \beta)$. If $W_4 = V$, then we are done with the proof. Otherwise, pick $w_4 \notin W_4$ and consider $\beta + iw_4, i = 1, \dots, 5$. In a similar way, there exists a vector $\gamma \in V$ that is not contained in any of the W 's and α, β and γ are linearly independent. Inductively, there exist a basis of V such that no vector in the basis belongs to subspaces W_1 and W_2 .

Problem 6

Write $y_i = v_i - v_{i-1}$, for $i \leq n-1, y_n = v_n$. We wish to prove that

$$\text{span}(v_1, \dots, v_n) = \text{span}(y_1, \dots, y_n)$$

since every y_i is a linear combination of the v_i , every expression $\sum a_i y_i$ may be rewritten as a sum $\sum \beta_i v_i$. Thus $\text{span}(y_1, \dots, y_n) \subset \text{span}(v_1, \dots, v_n)$.

On the other hand, we may write $v_n = y_n, v_{n-1} = y_{n-1} + y_n, \dots, v_1 = y_1 + \dots + y_n$. Hence, every v_i is a linear combination of the y_i , and $\text{span}(y_1, \dots, y_n) \subset \text{span}(v_1, \dots, v_n)$.

Therefore, $\text{span}(v_1, \dots, v_n) = \text{span}(y_1, \dots, y_n) = V$. Since $\dim(V) = n$, $\{y_1, \dots, y_n\}$ must be linearly independent as well, hence it is a basis.